Advanced techniques in applied economics

Lecture 8: Positive dependence and total positivity

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When the sign of dependence is itself useful structure

Why talk about positive dependence?

Most of the course has focused on sparsity, conditional independence, and latent structure. But another kind of useful structure is the **sign** of dependence.

Economic examples

- asset returns often comove positively,
- macro indicators may move together in expansions and recessions,
- symptoms of a syndrome are often all positively associated,
- a one-factor CAPM-style model gives positive comovement.

Statistical payoff

Sign constraints can act as regularization:

- existence of estimators improves,
- fitted graphs can become sparse,
- covariance estimation can be more stable.

Main message: knowing that dependence is positive is already a structural assumption — and sometimes a very powerful one.

Roadmap for today

1. positive association and MTP_2
2. the Gaussian and Ising special cases
3. why MTP_2 can be statistically useful
4. covariance estimation under total positivity
5. applications to finance and other data

Part 1: Positive dependence and MTP_2

Positive association

A random vector $X = (X_1, \ldots, X_m)$ is **positively associated** if for any two non-decreasing functions ϕ, ψ,

$$\mathrm{cov}\{\phi(X), \psi(X)\} \geq 0.$$

Interpretation

If one collection of coordinates is large, then other increasing functions of the data also tend to be large.

Examples

- positively correlated stock returns,
- symptoms of a common disease,
- variables driven by a common positive latent factor.

Multivariate total positivity of order two

A density f is MTP_2 if for all x, y,

$$f(x)f(y) \leq f(x \wedge y)\, f(x \vee y),$$

where $x \wedge y$ and $x \vee y$ denote componentwise minima and maxima.

Why this matters

MTP_2 is a strong and tractable form of positive dependence. It implies positive association and has powerful closure properties.

Basic properties

If X is MTP_2, then:

- every marginal distribution is MTP_2,
- every conditional distribution is MTP_2,
- and X is positively associated.

No Yule–Simpson paradox for signs

Positive dependence behaves differently from arbitrary dependence.

A useful intuition

With unrestricted models, marginal and conditional associations may change sign in messy ways. With MTP_2 models, signs are much more stable.

Why this is interesting

This means that positive dependence is not only a descriptive feature. It also simplifies the conditional independence and graphical structure.

Part 2: Gaussian and binary special cases

Two important special cases

Gaussian model:

$$X \sim N(0, \Sigma), \qquad K = \Sigma^{-1}.$$

Then

$$X \text{ is } \mathrm{MTP}_2 \qquad \Longleftrightarrow \qquad K_{ij} \leq 0 \text{ for all } i \neq j.$$

Binary Ising model:

$$p(x) \propto \exp(h^\top x + x^\top Jx/2).$$

Then

$$X \text{ is } \mathrm{MTP}_2 \qquad \Longleftrightarrow \qquad J_{ij} \geq 0 \text{ for all } i \neq j.$$

Gaussian MTP_2 means an M-matrix precision

In the Gaussian case, MTP_2 is equivalent to saying that the precision matrix is an **M-matrix**:

$$K_{ii} > 0, \qquad K_{ij} \leq 0 \;\; (i \neq j).$$

Consequences

This implies:

- all correlations are nonnegative,
- all partial correlations are nonnegative,
- the graph is an *attractive* Gaussian graphical model.

In the Gaussian world, total positivity is a sign constraint on the inverse covariance matrix.

Why this is less restrictive than it looks

At first sight, MTP_2 sounds severe. But in sparse models or latent-factor models it often appears naturally.

Examples where MTP_2 arises

- one-factor models with positive loadings,
- latent tree models,
- ferromagnetic Ising models,
- Brownian motion tree models.

Economic intuition

If a common hidden force pushes many variables in the same direction, positive dependence is natural.

Part 3: Why total positivity is statistically useful

Why MTP_2 is useful for estimation

The sign restriction is not just philosophically appealing. It improves the statistics.

Key facts in the Gaussian case

- the MLE under MTP_2 exists with only two observations,
- the fitted inverse covariance is sparse,
- the constraint acts like a built-in regularizer.

Why economists should care

This is highly relevant in finance and macro, where covariance estimation is hard and many variables are plausibly positively dependent.

A financial example

Monthly correlations among major stock-market indices are often all positive. A sample correlation matrix like

$$S = \begin{pmatrix} 1 & 0.61 & 0.73 \\ 0.61 & 1 & 0.55 \\ 0.73 & 0.55 & 1 \end{pmatrix}$$

already suggests a positive-dependence story.

Why this matters

If the fitted precision matrix also has non-positive off-diagonal entries, then the Gaussian model is MTP_2 and the whole dependence structure is being regularized by that sign information.

A one-factor finance story again

Suppose returns satisfy

$$X_i = \lambda_i F + \varepsilon_i, \qquad \lambda_i > 0,$$

with a common market factor F.

Implication

All cross-covariances are nonnegative:

$$\mathrm{cov}(X_i, X_j) = \lambda_i \lambda_j \,\mathrm{var}(F) \geq 0.$$

This is one reason positive dependence is natural in finance.

Message

If your economic story is "everything loads positively on the same latent factor", then MTP_2 is not exotic at all.

Part 4: Covariance estimation under total positivity

The estimation problem

Given a sample covariance matrix S, we want to estimate a covariance matrix under the constraint that the distribution is Gaussian MTP_2.

Equivalent formulation

Estimate a positive definite precision matrix K solving

$$\min_{K \succ 0} \left\{ -\log\det K + \operatorname{tr}(SK) \right\}$$

subject to

$$K_{ij} \leq 0 \qquad (i \neq j).$$

Interpretation

This is a sign-constrained Gaussian graphical model. The sign constraint replaces part of the usual tuning or regularization burden.

A Gaussian MTP_2 fit in R

A package route is to use golazo for sign-constrained Gaussian graphical models.

```r
# if needed:
# devtools::install_github(
# "pzwiernik/golazo")

library(golazo)
R <- cor(dat)
res <- positive.golazo(R, rho = Inf)
K <- res$K
```

```r
library(qgraph)
qgraph(-cov2cor(K),
       layout = "spring",
       threshold = 0.05)
```

Interpretation

The estimated precision matrix respects the positive-dependence sign constraints and typically becomes sparse.

A robust variation

For heavy-tailed financial data, one can combine total positivity with rank-based correlations.

```
kR <- cor(dat, method = "kendall")
sR <- sin(pi * kR / 2)
res <- positive.golazo(sR, rho = Inf)
K <- res$K
```

```
qgraph(-cov2cor(K),
       layout = "spring")
```

Interpretation

This is the non-paranormal idea from Lecture 2 combined with the positive-dependence constraint.

What recent work says

Recent work has pushed this in two directions:

- more efficient algorithms for large Gaussian MTP_2 graphical models,
- applications to covariance estimation and portfolio selection.

Examples

- adaptive estimation of Gaussian graphical models under total positivity,
- covariance estimation under total positivity for portfolio selection,
- local and extremal variants of positive dependence beyond the classical Gaussian setting.

Part 5: Binary and latent-variable examples

A binary latent-class example

Positive dependence also appears naturally in binary data.

Example

Suppose three symptoms are conditionally independent given a latent binary disease state H:

$$X_1 \perp\!\!\!\perp X_2 \perp\!\!\!\perp X_3 \mid H.$$

If the disease state raises the probability of all three symptoms, then the observed distribution is positively dependent — in fact often MTP_2.

Interpretation

A latent class model with a common severity variable often produces positive dependence among all observed symptoms.

Binary MTP_2 modeling in R

For binary data, the package `MTP2binary` provides one route to fitting MTP_2 models.

```r
# if needed:
# devtools::install_github(
# "pzwiernik/MTP2binary")

library(MTP2binary)
```

```r
# package functions depend on model
# class and data format;
# see package examples/vignettes
```

Teaching point

The binary and Gaussian cases are different models, but the same positive-dependence idea reappears in both.

Take-away

- Positive dependence is a real structural assumption, not just a descriptive comment.
- MTP_2 implies strong closure, monotonicity, and graphical properties.
- In Gaussian models, MTP_2 is equivalent to an M-matrix precision.
- This sign structure can be exploited for covariance and graph estimation.
- In finance and related applications, total positivity can be both economically plausible and statistically useful.

Sparsity is not the only useful structure in dependence modeling. The sign of dependence can be informative too.

Looking ahead

Final project direction

Many of the ideas in this course can be turned into projects:

- sparse Gaussian graphs,
- LiNGAM and non-Gaussian identification,
- hidden confounding and proxies,
- spillovers on networks,
- positive-dependence covariance estimation.

Final thought

A useful model is often one that encodes the right kind of structure: independence, sparsity, latentness, direction, or sign.